

Integrative Medicine in Rehabilitation

AJIT B. PAI, LORI V. SHUART, AND DAVID F. DRAKE

SUMMARY PEARLS

- Integrative health focuses on a holistic, patient-centered approach to health and wellness and is widely used in behavioral health, pain management, and neurorehabilitation.
- Individuals can benefit from singular or multiple mind-body practices integrated into traditional health care.
- Nutritional supplements and essential oils can interact with medication.
- The rising cost of health care creates opportunities to utilize an integrative approach to improve quality of care and reduce cost.

Introduction

Integrative health care focuses on a holistic, interdisciplinary, patient-centric, whole health approach to health and wellness. There is emphasis on prevention of disease, not just treatment of diagnosed conditions. In addition, multiple aspects of the person are treated at once, sometimes referred to as the mind, body, and spirit. Over the years, the use of integrative approaches has grown across the United States. A 2022 national health interview survey (Table 19.1) found a significant increase in use of complementary and integrative health (CIH) modalities compared with 2002.⁸⁹ In addition, 27% of US military veterans use CIH modalities in the treatment of chronic musculoskeletal pain.¹¹⁶ Integrative health care takes into consideration the whole person and all things relating to the person's life. This often includes empowering individuals to actively participate in their own health and well-being by providing education and resources, but most importantly it involves active listening to glean an understanding of their own goals. Doing so requires care providers to undergo a cultural change whereby they no longer consider health care to be problem focused but rather ascribe to the philosophy of integrative health care. This reduces the reactive nature of diagnosis and treatment in modern medicine. Instead, providers include discussions about goal attainment for prevention of disease. These discussions and direction help individuals to make long-lasting changes that proactively promote health.

Oftentimes, providers do not have the opportunity to spend enough time with the patient. It is at this point that an integrative health coach (IHC) may be useful. These care team members assist individuals with goal setting and progress tracking. IHCs are formally trained care providers who meet individually with patients to help promote health and wellness. They often use resources such as the Circle of Health (Fig. 19.1). Exploring all components of individuals' health is key to making lifelong changes. When a care provider or care team assists with implementing lifelong changes

that mirror the individuals' goals to improve health and wellness, then a major accomplishment is achieved.

Whole Medical Systems

Integrative health care incorporates practices and philosophies from whole medical systems into conventional or Western medicine. Over the centuries, there have been many different types of whole medical systems. Each system incorporates a primary philosophy and includes diagnosis and treatment of conditions. In this section, we highlight some major whole medical systems.

Traditional Chinese Medicine

Traditional Chinese medicine (TCM) is probably the best known of the whole medical systems. It uses specific diagnostic evaluations such as pulse and tongue assessment. Treatments include herbal prescriptions and interventions such as acupuncture, massage, exercise, dietary modifications, and meditation. This practice, rooted in Taoist philosophy, evolved over thousands of years in China. This philosophy uses the concept of balance between complementary forces called yin and yang. An imbalance between the two will result in disease. While most research is done on the specific modalities, a recent study evaluated TCM rehabilitation, which included TCM evaluation with tuina and scalp acupuncture. They found improvement in spasticity and activities of daily living.¹³⁷

Ayurveda

Ayurveda is a whole medical system from India that aims to provide guidance regarding food and lifestyle to either maintain wellness or improve health. It understands that there are energetic forces (tridoshas) that influence nature and human beings and that there is a strong connection between the mind and the body.

TABLE 19.1 Percent of US Adults Using Integrative Approaches

Approach	2022	2024
Acupuncture	1%	2.2%
Chiropractic	7.4%	11%
Guided imagery/Progressive muscle relaxation	3.8%	6.4%
Massage therapy	4.8%	10.9%
Meditation	7.5%	17.3%
Naturopathy	0.2%	1.3%
Yoga	5%	15.8%

Data from Nahin RL, Rhee A, Stussman B. Use of complementary health approaches overall and for pain management by US adults. *JAMA*. 2024;331(7):613-615.

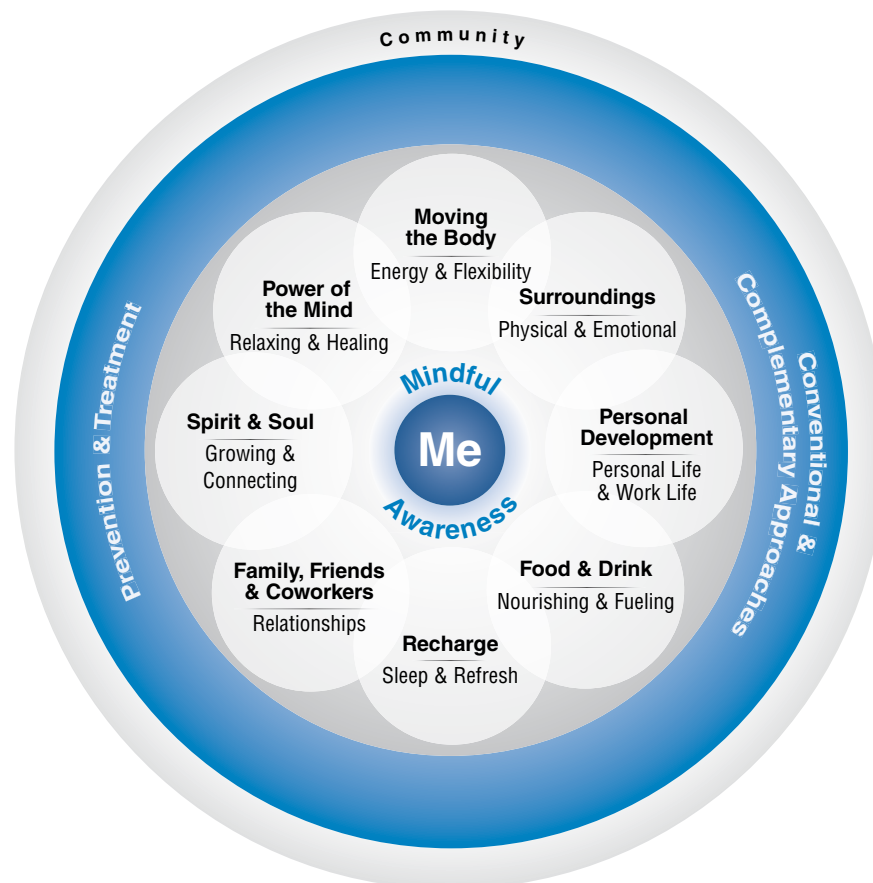
Dietary changes along with herbs are often the prescribed remedy. Although research on Ayurveda is sparse, a systematic review found Ayurveda management, including Panchakarma therapy and Ayurveda medications, with other alternative therapies, such as acupuncture, yoga, and rehabilitation, improved muscle strength, quality of life, range of motion, and neuronal function. It reduced depression, stress, and pain with symptom scores.⁶³

Homeopathy

Homeopathy, developed in Germany in the late 18th century, focuses on two theories: “like cures like” (i.e., the disease can be treated or cured by a substance that produces similar symptoms in healthy people) and “law of minimum dose” (i.e., the effect is greater at the lowest dose of the medication). Many homeopathic remedies are so diluted that no molecules of the original substance remain.

Naturopathy

Naturopathy, or naturopathic medicine, is a medical system that emerged from 19th-century Europe that uses a combination of traditional practices and health care approaches popular at the time. Naturopathic practitioners see people for various health-related concerns, including primary care, overall well-being, and treatment of illnesses. Treatment approaches include dietary and lifestyle changes, stress reduction, herbs and other dietary supplements, and homeopathy, manual therapies, and exercises. They may work with or refer patients to conventional health care providers. Most research does not consider naturopathy as a separate intervention from conventional medical practices; however, in 2019, a systematic review looking at a broad range of diagnoses concluded naturopathic multimodality therapies were shown to be effective in the treatment of cardiovascular disease, musculoskeletal pain, type 2 diabetes mellitus, polycystic ovary syndrome, depression, anxiety, and other chronic complex conditions.⁸⁸



• **Fig. 19.1** Components of proactive health and well-being. (From Veterans Health Administration. Office of Patient Centered Care and Cultural Transformation.)

Functional Medicine

Functional medicine determines how and why illness occurs and restores health by addressing the root causes of disease for each individual. It is an individualized, patient-centered, science-based approach that empowers patients and encourages a patient-practitioner relationship, working together, to address the underlying causes of disease and promote optimal wellness. A detailed understanding of each patient's genetic, biochemical, and lifestyle helps to direct a personalized treatment plan that leads to improved patient outcomes.

Mind-Body Practices

According to the National Center for Complementary and Integrative Health (NCCIH), mind-body practices are “techniques that strengthen the connection between the emotional, mental, and physical aspects of ourselves.” Mind-body skills are built on the understanding of neuroplasticity. The more you practice, the more you build neurologic pathways between the body and mind and thus create a stronger connection that can change the structure and function of the brain.

Practices of the Mind

Meditation and Mindfulness

Meditation implies a variety of practices that focus on mind-body integration and are used to calm the mind and enhance overall well-being. Meditation is a frequently used modality to induce relaxation and reduce stress. The primary goal of meditation is to focus attention to control thoughts and reduce stress. An added benefit is that no special equipment or space is necessary; this makes it accessible to anyone. Meditation may involve maintaining mental focus on a particular sensation, such as a sound, a visual image, breath, or a mantra, which is a repeated word or phrase. Other forms of meditation may include the practice of mindfulness, which involves maintaining attention or awareness on the present moment without making judgments. The most common form of meditation is prayer.

In physical medicine and rehabilitation practice, mindfulness meditation is often used with individuals who suffer pain or experienced life-altering traumatic injuries. In a seminal article in 1982, Jon Kabat-Zinn demonstrated that mindfulness significantly improved pain rating in a majority of participants along with mood and other psychiatric symptoms.⁶⁰ More recently, mindfulness meditation was associated with pain reduction in patients with chronic low back pain^{15,33,57} and fibromyalgia¹¹⁴ along with significant improvement in depression, physical health-related quality of life, and mental health-related quality of life.⁴⁹ The effects of mindfulness can be conducted in-person or via telemedicine with similarly effective results.³² In addition, for individuals with traumatic brain injury (TBI), mindfulness meditation was shown to improve quality of life with sustained improvement beyond 1 year.⁶

Guided Imagery

Guided imagery, or visualization, includes picturing objects, scenes, or events that are associated with relaxation, calmness, or movement and attempting to produce similar feelings in the body. The American Holistic Nurses Association defines guided imagery as a modality that “assists clients in connecting with their inner knowledge at the thinking, feeling, and sensing levels, promoting their

innate healing abilities. Together, guide and client cocreate an effective way to work with: pain, symptom grief, and stress management; conflict resolution; self-empowerment issues; and/or preparing for medical-surgical procedures.”⁴⁴ According to the Academy for Guided Imagery, there are three categories of therapeutic application for guided imagery: (1) stress reduction and relaxation, (2) active visualization or directed imagery, and (3) receptive imagery.

In the rehabilitation world, motor imagery, a subset of active visualization or directive imagery, is used to prepare individuals for functional movement. Guided imagery reduces overall pain, phantom limb pain in amputees,⁷⁵ and pain in advanced cancer patients. In addition, there is evidence of reduction in anxiety,⁶⁶ length of stay in postoperative orthopedic patients,²⁰ and improvement in frequency, severity, quality of life, and disability caused by chronic tension-type headache.⁸⁰

Mindfulness-Based Stress Reduction

Mindfulness-based stress reduction (MBSR) is intensive mindfulness training, including meditation, yoga, body awareness, behavioral awareness, and emotional awareness. It explores an individual's present experience in relation to current thoughts, physical and emotional sensations, and memories to promote understanding, acceptance, and reduction of suffering. In adults with chronic low back pain, treatment with MBSR compared with the control group demonstrated greater improvement in back pain and functional limitations at both 26 and 52 weeks.²³ However, in individuals with Parkinson disease, there is inconclusive evidence for the use of MBSR.⁸² Veterans with posttraumatic stress disorder (PTSD) enrolled in the MBSR program showed a reduction in PTSD symptoms and specifically showed increases in “acting with awareness” and “nonreactivity.” They also noted increases in mindfulness were strongly related to decreases in hyperarousal and emotional numbing.¹¹²

Hypnosis

Clinical hypnosis is a heightened state of concentration and focused attention. Guided only by a trained professional, hypnosis enables the recipient to be more open to suggestions to make changes in perceptions, emotions, sensations, or behaviors to lead toward better health. Recently, more focus has been placed upon hypnosis for the use of pain control. In 2018, hypnosis was found to provide a statistically significant improvement in pain and anxiety intensity for individuals undergoing wound care to treat burns.⁹⁸ At this time, a wide area of research in hypnosis involves palliative care. However, there is no conclusive evidence that hypnosis is a primary treatment modality for any medical condition.

Prayer

Prayer is the best known and most widely practiced example of meditation. It is used with the intention of health promotion and alleviation of disease. Traditional prayer allows for a deep state of meditation. Intercessory prayer is praying for others. Currently, there are inconclusive results as to the effect of prayer on illness. One meta-analysis of studies examining the effect of private and communal prayer participation suggests that prayer might reduce pain intensity.⁵⁶

Creative Arts Therapies

Creative arts therapies (CATs) include art therapy, music therapy, dance/movement therapy, and many other forms of expression. These modalities are used to help individuals with illness or injury

to use expression and communication to improve their functioning. In persons with stroke, CAT in addition to physical therapy significantly decreased depression and improved physical function and quality of life compared with physical therapy alone.⁶⁵ In addition, music therapy may improve gait, upper extremity function, communication, and quality of life.⁷⁹ For individuals on mechanical ventilators, music therapy may have a beneficial effect on anxiety, decrease consumption of sedative and analgesic medication, and reduce respiratory rate and blood pressure.¹⁴

The previously discussed examples are just a sampling of the research that exists for practices of the mind. Overall, meditation and CATs can be used in many clinical areas to assist with adjustment, pain symptoms, frustration, and more. The incorporation of this practice into rehabilitation programs can potentially assist individuals with their recovery. However, more research is needed to truly demonstrate the effectiveness of these modalities in a variety of conditions that cause functional limitation.

Movement Therapies

Exercise and Mindful Movement

Exercise has been repeatedly shown to reduce the risk for development and progression of diseases, illnesses, and mortality. Physical activity is applied in recovery of illnesses and deconditioning; in addition, it is recognized to provide medicinal properties in the preventative and wellness world. In 2007, the American College of Sports Medicine collaborated with the American Medical Association and the Office of the Surgeon General to implement the Exercise Is Medicine (EIM) campaign.¹¹⁸ The EIM initiative was implemented to encourage practitioners to promote exercise or physical activity for treatment of disease and preventative health measures.

Although many types of aerobic and anaerobic exercise promote physical and mental well-being, there are specific types of physical activity that incorporate a mind-body approach. These practices are often referred to as mindful movement. Mindful movement is any application of combined movement of physical-based activity, mindfulness practice, breathing, and spiritual and/or energy-generating practice. Mindful movement aids in symptomatic management of depression, schizophrenia, and other mental health and psychiatric disorders.^{72,139} There is also a strong relationship between biopsychosocial improvements and mindful movement to include enhanced emotional well-being and improved social relationships.¹¹ Although there are many types of mindful movement, only a select few are supported by evidence-based research to justify their use in medicine. The hormone irisin is released from muscle tissue during exercise; recent studies show this hormone is also released with forms of tai chi, and it can have a significant effect on cognitive outcomes with aging adults.⁴²

In response to the need for nonpharmacologic strategies to address chronic low back pain and other chronic pain conditions, a mixed method feasibility trial was conducted integrating group therapy, mindfulness, and expressive dance and movement.¹⁰⁹ In this study, maladaptive pain-related beliefs and coping strategies were challenged, and the results demonstrate a strong indication for chronic pain management and increased motivation for physical activity. Although more research is warranted, the results for mindful movement to address chronic pain are very promising.

In a 2024 systematic literature review conducted over a 10-year period, a combination of exercise, cognitive behavioral

therapy, and mindfulness-based programs showed a significant reduction in pain and fatigue during cancer treatments.¹³ Specifically, tai chi and qigong proved beneficial during active oncology treatment, whereas yoga had more benefit managing symptoms after treatments.

Tai Chi

Tai chi (pronounced “tie-chee”) is a form of martial arts derived from ancient traditional Chinese practice.⁴⁷ Tai chi involves a series of movements performed in a slow manner accompanied by deep or patterned breathing. Tai chi fosters qi (pronounced “chee”), or life energy, within the body with efforts to promote harmony of mind and body. As with many martial arts, there are several forms of tai chi practiced. The most common forms of tai chi include chen, yang, wu, and sun. All styles share the same essential principles but have their own unique characteristics. The basic principle of all forms or styles is to integrate the controlled breathing with movement, generate energy, and harvest strength and balance for both the mind and body.

There have been several evidence-based health benefits researched to support the use of tai chi in practice.⁵³ The most prominent benefits for the application of tai chi are for (1) fall prevention,^{106,126} (2) osteoarthritis,¹³⁰ (3) Parkinson disease, (4) chronic obstructive pulmonary disease (COPD),²¹ and (5) cognitive function.¹³⁶ Specifically, the use of tai chi with a combination of various interventions for individuals with Parkinson disease has been shown to improve motor function, depression, quality of life,¹¹¹ balance and well-being,³¹ and functional fitness.²⁴ Ai chi, a tai chi application in an aquatic setting, may help to improve functional mobility, quality of life, balance, and motor ability more efficiently than similar land applications for those with Parkinson disease.⁶⁷ There is also good evidence supporting health benefits to those with depression,¹²⁴ cardiovascular disease,⁶⁸ cancer,¹³⁶ fibromyalgia,¹⁰⁷ PTSD and other similar anxieties,⁶² hypertension,⁹⁴ and osteoporosis.²⁵ Recent research supports the use of tai chi in reducing dyspnea in patients with lung cancer undergoing rehabilitation.¹¹³

Qigong

Qigong (pronounced “chee-gung”) is another traditional Chinese practice of movement encompassing affirmations, breath work, meditation, and guided imagery or visualization. *Qi* means “air or breath,” and *gong* means “effort or work.” In the practice of qigong, the breath is the vital life force that controls everything in the universe. This belief correlates with the concept of bioelectricity through the body. Qigong, when put into practice, requires time, patience, and repetition. It is believed that individuals will develop the ability to control the qi and prevent disease, promote self-healing, and increase longevity.

Qigong is proven to be effective for improving exercise capacity for individuals who are deconditioned, decreasing symptoms of depression,²² reducing inflammation,⁹⁷ stabilizing blood sugar,⁸⁵ reducing musculoskeletal pain, improving quality of life, and improving quality of sleep.¹³⁸ Paired with other traditional therapies, qigong may decrease recovery times^{77,84} and increase pulmonary function for persons with COPD.⁷³

A recent systematic metaanalysis identified a variety of exercise- and mindfulness-based interventions for impact on cognition and quality of life in breast cancer survivors during and after cancer treatment.¹³⁴ This analysis concluded that physical activity and qigong improved executive functions over other interventions. Mindfulness-based movement practices are proving to be an effective intervention in cancer rehabilitation programs.

Yoga

Yoga is a mindful movement practice that encompasses stretching, strengthening poses, meditation, and breathing exercises. Yoga uses body, breath, and the senses to reconnect the participant with the universe and move emotions and thoughts into stillness. The most common forms of yoga include hatha, vinyasa, ashtanga, bikram, iyengar, kundalini, yin, viniyoga, anusara, and sivananda. Hatha yoga is the most beginner-friendly yoga and is often integrated into rehabilitative practice. Yoga decreases chronic pain, stabilizes blood pressure, and reduces stress. Supportive data also suggest that yoga improves balance, breathing, spine health, mental health, blood flow, sleep, strength, and flexibility. Currently, there is some evidence that the practice of yoga results in improvements in low back pain at 3 and 6 months as compared with nonexercise controls.¹²⁵ However, in persons with stroke, there is no conclusive evidence that yoga is effective or safe.⁷⁰

In addition, yoga provides positive effects on the cardiac autonomic nervous system while reducing inflammation, decreasing stress, and reducing the risk and progression of cardiovascular disease.⁴³ There is relatively strong evidence that supports the practice of yoga for low back pain and other chronic pain syndromes as a long-term intervention and effect on disability.²⁸ Yoga is associated with improved regulation of the sympathetic nervous system and the hypothalamic-pituitary system in several populations and therefore associated with symptoms of stress reduction.⁹⁶ Yoga increases range of motion and mobility of the spine; consequently, individuals report improved quality of life.⁴¹

Pilates

Pilates is a form of mindful movement that was developed by Joseph Pilates in the 1920s. He developed the method, once called *contrology* (the science of control), incorporating components of gymnastics, boxing, and wrestling. The origin of Pilates was for the purpose of rehabilitating soldiers returning home from war. Pilates practice emphasizes the development of core strength and spinal alignment to improve overall fitness through the coordination of body, mind, and spirit. Pilates can be performed independently or in a group, on a mat, or using specialized equipment. Some specialized equipment used in Pilates include the Reformer, the Cadillac, and the Wunda chair, using a system of pulleys and straps to provide support and resistance.

Regular practice of Pilates can improve posture, muscle tone, balance, and joint mobility and relieve stress and tension. For elite athletes, including dancers, Pilates can complement training by developing whole body strength and flexibility and can help to reduce the risk of injury. Pilates is used as an effective rehabilitation tool, supplementing traditional applications, primarily in reducing pain and disability.¹⁸

Pilates has been proven to be effective in treatment for low back pain by improving the lumbopelvic rhythm⁸¹ and strengthening the spinal muscles to improve neuromuscular efficiency⁹⁵ and core muscles, including the lower pelvic floor.³⁸ Pilates is useful with patients with chronic neck or back pain as an effective tool in the management of pain, disability, and kinesiophobia.³⁰ It improves overall function and quality of life and leads to reduction in the use analgesic pain medication.³⁴ In rehabilitation, Pilates is more effective than closed kinematic chain exercises for reducing pain and improving functional performance with subjects with knee osteoarthritis.^{83,110}

In the older adult population, Pilates is used to improve lung function and physical conditioning.² Overall, improvement in exercise tolerance is seen with clients with chronic diseases using

regular Pilates.⁸⁶ Pilates may be beneficial to use as an alternative to standard physical activity applications in reducing depression and anxiety.¹²³ Pilates, like other forms of exercise, reduces the fall risk of older adults by improving static and dynamic balance, range of motion, and functional mobility.¹⁰³

Neurologic Pilates was developed as a modified version of the Pilates method to be used for neurologic rehabilitation. Recent findings indicate that, over a 3-month period, the integration of neurologic Pilates significantly improved static balance, righthand manual dexterity, and daily functioning of stroke survivors when combined with conventional therapies; however, scientific evidence remains limited.³⁹

Feldenkrais Method

The Feldenkrais method or therapy, developed by Moshe Feldenkrais, is the application of a gentle mindful movement based on principles of physics, biomechanics, psychology, motor development, martial arts, and the reconnection of self-education to how the body moves and develops. Feldenkrais stated, "We move according to our perceived self-image." By expanding clients' perception and increasing awareness, the theory is that participants will become more aware of their habits and tensions and develop new ways of moving. Range of motion, flexibility, and coordination become easier, and movement thus becomes more efficient.

There are two general types of Feldenkrais practices: (1) awareness through movement and (2) functional integration sessions. Awareness through movement uses slow, mindful movements to improve strength, flexibility, and integration of the body and mind. Somatic education is integrated into every session to relate movement to functional daily activities. Functional integration sessions use gentle touch to guide the participant through movement patterns enhancing awareness and improving ease and effectiveness of movement. The principles of body awareness and self-discovery are practiced and integrated into Feldenkrais sessions.

Recently, Feldenkrais has been shown to (1) improve efficiency of physical movements and self-efficacy⁴⁸; (2) produce changes in muscle tone leading to a more relaxed spine and reduction of chronic pain¹⁶; (3) stabilize balance,¹¹⁹ improve quality of life,¹¹⁷ and promote mobility and confidence⁹³; and (4) offset age-related declines and enhance cognitive function.¹²¹ In addition, individuals with chronic pain benefit from Feldenkrais due to changes in muscle tone, improvements in mobility and balance, and improved confidence in movement through awareness.⁹¹ With Feldenkrais, older adults experience improvements in balance, mobility, gait, and fall reduction.¹²² A systematic review meta-analysis of the Feldenkrais method indicated that it is an effective treatment of some clinical conditions and pathologies by improving balance, mobility, and quality of life and showing similar effects as conventional interventions.⁷

Biomechanical Therapies

Acupuncture

Acupuncture is a form of energy medicine and is one of the more common and more researched of the integrative health modalities. It shares with the others a similar treatment setting and a conceptual framework similar to tai chi or qigong, with a life energy called *qi* that is thought to circulate through all parts of the body in energy channels called meridians. These meridians connect the exterior to the interior and the organs to each other and the exterior. The classical Chinese explanation is that channels of

energy run in regular patterns through the body and over its surface. These energy channels (meridians) are like rivers flowing through the body to irrigate and nourish the tissues. An obstruction in the movement of these energy rivers is like a dam that backs up in others. Pain and illness are thought to occur when the flow of qi becomes blocked or unbalanced. Acupuncture is a treatment used to reestablish the flow of qi through the placement of needles at points along the meridians, thus allowing the body to return to homeostasis and easing the ailment for which it was prescribed. The meridians can be influenced by needling the acupuncture points. The acupuncture needles unblock the obstructions at the dams and reestablish the regular flow through the meridians. Acupuncture treatments can therefore help the body's internal organs to correct imbalances in digestion, absorption, and energy production activities and in the circulation of energy through the meridians.

The modern scientific explanation is that needling the acupuncture points stimulates the nervous system to release chemicals in the muscles, spinal cord, and brain. These chemicals will either change the experience of pain or will trigger the release of other chemicals and hormones, which influence the body's own internal regulating system. An action potential is also triggered by the stimulation of acupuncture needle insertion, which in turn affects the activities in the brain's neuronal network.¹³¹ The physiologic reactions of acupuncture effects are beginning to emerge. Stimulation of acupuncture points has been shown to create signal changes in the amygdala, anterior hippocampus, and subgenual cingulate cortex on functional magnetic resonance imaging (fMRI).⁷⁶ Other studies confirm these findings, defining the role of the amygdala in affect, fear, and defensive behavior, as well as the processing of pain^{8,12,55} and motivation.¹³³ The hippocampus is thought to link affective states with memory processing. The signal decrease within the amygdala and anterior hippocampus is consistent with past acupuncture fMRI studies at acupoints LI-4 and GB-34, as well as ST-36.^{51,52,128,129,135} Ahsin et al. linked electroacupuncture to functional improvement by the measured rise in endorphins and fall in cortisol levels in individuals given electroacupuncture.¹

The improved energy and biochemical balance produced by acupuncture results in stimulating the body's natural healing abilities and in promoting physical and emotional well-being. The anatomy of acupuncture points is not clear; however, a theory proposed by Langevin and Yandow⁶⁹ suggests that the meridians are located within tissues planes. Acupuncture points occur at a convergence of these planes, and meridian qi is the biochemical and bioelectrical signaling in the connective tissue. A blockage of qi may lead to an altered connective tissue matrix composition, leading to altered signal transduction and therefore pain or other symptoms. They reported that ultrasound imaging reveals visible connective tissue forming an intramuscular cleavage plane at acupuncture points but not at control points.⁶⁹ Another reported a distinctive feature of acupuncture points is the high density of nerve endings.^{46,71} Dorsher did a study based on Travell's and Simons's texts and found that 93.3% of the common myofascial trigger points (238 of 255) correspond to the classical acupuncture points.³⁵ The fascial anatomy has a striking anatomic correlation to acupuncture points and meridians. Finando and Finando suggested that all fundamental characteristics of acupuncture treatment are consistent with stimulation of the fascia.³⁷

In 2022, an evidence map reviewing acupuncture was published and identified 1207 potentially relevant citations and included 64 publications, categorized into more than 40 conditions.

Their general observations include (1) there are more mapped conclusions for painful conditions than for all other conditions combined, (2) there are only a small number of reviews with at least one conclusion rated as high certainty of evidence (CoE), (3) the conclusions about the effectiveness of acupuncture that were judged to have at least moderate CoE mostly concerned comparisons of acupuncture to sham or control acupuncture for painful conditions or pain outcomes, and (4) there is no evidence that acupuncture is less safe than usual care for these conditions.¹⁰⁸ Acupuncture is an ancient medical modality, with physiologic changes that appear to suggest neurologic and neurochemical effects that can lead to clinical improvement. Many people may benefit from acupuncture, although it is not a panacea.

Osteopathy

Since its founding in the 19th century by Andrew Taylor Still, osteopathic medicine, which includes manual manipulation of the spine, has been at the forefront of integrative medicine. Still stressed the importance of the connection among mental, physical, emotional, and spiritual health and taught that each plays an important role in patients' overall health. The evidence for manual therapy, as many refer to it, is not overwhelming. However, there does appear to be a role for osteopathic medicine in the management of pain, particularly if treated early. A 2018 systematic review and metaanalysis looking at manipulation and mobilization for chronic low back pain, sponsored by the NCCIH, determined there is moderate-quality evidence that manipulation and mobilization are likely to reduce pain and improve function for patients with chronic low back pain; manipulation appeared to produce a larger effect than mobilization.²⁶

Chiropractic

Chiropractic considers the structure of the body and the relationship to function. By using manipulation of the spine or extremities, alignment is corrected so that pain and function can improve. Oftentimes, chiropractors use nonmanipulation modalities as part of their practice. These include heat and ice, electrical stimulation, nutritional counseling, relaxation techniques, and exercise. Chiropractic is often used for spinal pain, headaches, joint pain, and soft tissue injuries.

Massage

Massage is a general term for pressing, rubbing, and manipulating the skin, muscles, tendons, and ligaments to aid in relaxation and recovery from injury. Massage therapists typically use their hands and fingers but may also use their forearms, elbows, and even feet. Swedish massage is primarily used for relaxation and is most common in spa settings. Thai massage uses assisted stretching. Shiatsu is a Japanese technique in which finger pressure, often along meridians that may correspond with acupuncture points, is used. Deep tissue massage is used to target trigger points and release chronic muscle tension. Sports massage can be used before events as part of a warmup, during the event in cases of cramping, and after the event as part of the cooldown. Heat can be added to any of the above mentioned treatments via warmed stones, topical ointments, creams, or other emollients to aid in relaxation.

A systematic review in 2021 that included 18 randomized controlled trials (RCTs) comparing Chinese massage (tuina) plus conventional physiotherapy with conventional physiotherapy was completed. They reported improvements in upper limb motor function using the Fugl-Meyer Assessment and upper and lower limb spasticity using the modified Ashworth scale.

They concluded that therapeutic massage, especially tuina, in addition to conventional therapy is effective for improving motor function and for reducing spasticity in stroke survivors.¹⁹ In 2019, an evidence map, which identified 32 high-quality systematic reviews of massage to treat pain, concluded that there was low strength of evidence of potential benefits of massage for labor, shoulder, neck, low back, cancer, arthritis, postoperative, delayed-onset muscle soreness, and musculoskeletal pain. They reported that the low strength of evidence could be attributed to the fact that few primary studies of large samples with rigorous methods had been conducted, which leaves evidence gaps about specific massage types for specific pain. In addition, they mentioned that primary studies often do not provide adequate details of massage therapy, limiting the extent to which reviews are able to draw conclusions about characteristics such as provider type.⁸⁷ A systematic review of massage in the general pain population looked at 60 high-quality and 7 low-quality studies and found that massage therapy effectively treats pain compared with sham, no treatment, and active controls. Massage therapy was also beneficial for treating anxiety and found to improve health-related quality of life.²⁹

Rolfing

Rolfing, also called Rolf therapy or structural integration, is a holistic system of bodywork that uses deep manipulation of the body's soft tissue to realign and balance the body's myofascial structure. There is a paucity of research; however, one study showed improved gait in children with spastic cerebral palsy.⁷⁸ Another showed improvement in body image and low back pain following three structural integration sessions.⁵

Biofeedback

Biofeedback is a technique of making unconscious or involuntary bodily processes (such as heartbeats or brain waves) perceptible to the senses with the hope of manipulating them by conscious mental control. A 2019 evidence map that included 16 good-quality systematic reviews examined biofeedback alone or as an adjunctive intervention. They found clear, consistent evidence suggesting that biofeedback is effective for reducing headache pain and improving urinary incontinence after prostatectomy. It may provide benefit for fecal incontinence, balance, and stroke recovery.⁶⁴

Regenerative Medicine

Regenerative medicine is a branch of medicine concerned with developing therapies that regenerate or replace injured, diseased, or defective cells, tissues, or organs to restore or establish function and structure. Regenerative medicine treatments include stem cell therapy to stimulate tissue repair and regeneration at the site of damage or the transplantation of tissues or organs made or grown outside the body. A meta-analysis of 78 RCTs reported that a reduction in pain was associated with platelet-rich plasma (PRP) at 3 months and sustained until 1 year. More specifically, they mention low- to moderate-quality evidence supporting a reduction in pain for lateral epicondylitis and knee osteoarthritis at 1 year, noting that PRP characteristics did not influence results. They concluded that PRP leads to a reduction in pain; however, evidence for clinically significant efficacy is limited. The available evidence currently supports the use of PRP in the management of lateral epicondylitis as well as knee osteoarthritis.⁵⁸ A 2020 meta-analysis of the effect of PRP on knee lesions selected only 83 clinical trials using the revised Cochrane risk-of-bias tool to evaluate risk. They reported significant differences in favor of PRP

when compared to the control groups both on the recovery of knee disorders and during knee surgery.¹²⁰ More recently, there has been interest in mesenchymal stem cells as a form of regenerative medicine. While this is a growing area of research, there is no consistent strong research supporting its use.

Natural Products

Nutrition

Appropriate nutrition is a large part of CIH. There is a concern that various foods cause inflammation that can lead to disease states. One thing to consider is whether an antiinflammatory diet is proper for the individual patient. Eicosanoids are breakdown products of fatty acids. One product of the breakdown of omega-3 fatty acids is antiinflammatory prostaglandin; however, omega-6 fatty acids result in breakdown products that include proinflammatory prostaglandins. Omega-3 fatty acids can be found in coldwater fish, such as salmon, sardines, herring, and mackerel; ground flax seeds or flax seed oil; leafy green vegetables; and walnuts. Omega-6 fatty acids are found in red meats, dairy products, partially hydrogenated oils, and foods with long shelf lives.

Another area of concern is excess free radical production, which may lead to tissue damage and inflammation. It is thought that food highly rich in antioxidants may help to protect against free radical-induced damage.

Dietary Supplements

The dietary supplement (Table 19.2) industry is continually growing. Dietary supplement use increased between 2007 and 2018 due to an increase in single-nutrient products, despite a relative decrease in multivitamin use.²⁷ In rehabilitation literature, there are multiple studies reviewing the use of dietary supplements in various populations. In multiple sclerosis, for example, RCTs found the use of polyunsaturated fatty acids (omega-6 or -3) had no statistically significant effects on clinical outcomes.³

Persons with spinal cord injury (SCI) use dietary supplementation at a greater rate than the general population.⁹² For years in the SCI population, individuals tried cranberry supplementation as a preventive mechanism for urinary tract infection (UTI). A 2017 review of the effectiveness of dietary supplements in the SCI population demonstrates there is high-quality, conclusive evidence against the use of cranberry supplementation for prevention of UTIs.⁹⁰ In addition, this study reported moderate-quality evidence for the beneficial effect of vitamin D (increased bone mineral density in nonsmokers), alpha-lipoic acid (fasting blood sugar, waist circumference, and blood pressure), and omega-3 supplementation.

In persons with brain injury, dietary supplements are used increasingly. Omega-3 acids are thought to prevent axonal loss and are strong antiinflammatory agents.⁴⁵ Due to these two mechanisms, there is hope that omega-3 acids can be used prophylactically in athletes and the military; however, conclusive evidence does yet not exist. Another supplement to be considered in TBI is zinc. This essential trace element is linked to various neurologic disorders and may have promise in the treatment of TBI. One study demonstrated improvements in Glasgow Coma Scale scores as compared with controls.¹³²

Vitamin D is a fat-soluble steroid hormone that is synthesized from 7-dehydrocholesterol through a multistep conversion process

TABLE 19.2 Dietary Supplements

Name	Background	Common Use	Safety	Research
Acai	Fruit from the acai palm tree	Weight loss and antiaging	Drinking unprocessed acai juice is linked to Chagas disease; may affect MRI results	May help with reducing blood sugar and cholesterol in overweight people
Aloe vera	Use dates back >6000 years in Egypt from the aloe plant	Topical: remedy for skin conditions (acne, burns, lichen planus, psoriasis, cold sores) Oral: treatment for weight loss, diabetes, hepatitis, and inflammatory bowel diseases	Topical use is likely safe; oral use has been linked to carcinogenic activity and few reports of acute hepatitis	Aloe latex has strong laxative components; some evidence for the topical use of aloe for psoriasis and rashes
Asian ginseng	Use dates back >2000 years; believed to replenish energy in TCM	Used to improve general well-being, stamina, concentration, immune function; slow aging; relieve various health problems	Short-term use is considered safe; side effects include headache, sleep disorder, and digestive problems; may affect blood sugar	May affect blood pressure and blood sugar
Bitter orange	Used for constipation, nausea, and indigestion	Currently used for heartburn, loss of appetite, nasal congestion, and weight loss; sometimes applied to painful areas, bruises, and sores	Reports of fainting, myocardial infarction, and stroke	Oil may help with ringworm, tinea cruris, and tinea pedis
Bromelain	Mixture of enzymes from pineapple plant used for digestive disorders	Used for nasal swelling and inflammation, osteoarthritis, digestive issues, and muscle soreness	Reports of GI problems, tachycardia, and menstrual issues	Helpful for acute nasal and sinus inflammation; conflicting evidence for osteoarthritis and muscle soreness
Butterbur	Shrub that grows in wet, marshy areas; traditionally used for plague and fever and used to treat cough and asthma	Used to treat migraines and urinary tract issues	Unprocessed butterbur contains chemicals that can cause liver damage; side effects include headache, itchy eyes, diarrhea, drowsiness, and fatigue	AAN and AHS endorse the use to reduce migraine frequency
Chamomile	Two types: German and Roman; important medicinal herb in Egypt, Greece, and Rome	Used orally for insomnia, anxiety, and GI issues	Some reports of allergic reactions; interactions with cyclosporine and warfarin have been reported	May help with generalized anxiety disorder
Cranberry	Native of North America	Used as a supplement to treat UTIs	May cause GI problems or increase risk of kidney stones; may interact with warfarin	Mixed evidence of its use in various populations but conclusive evidence against its use in prevention of UTIs in SCI
Echinacea	Nine known species; used by Native Americans as traditional medicines	Used for treating the common cold and thought to improve immune system function	Most common side effects include nausea and stomach pain	Has not been proven to reduce time of illness; may slightly reduce the chance of catching the common cold
Evening primrose oil	Native Americans used for bruises, hemorrhoids, and minor wounds	Used for eczema, rheumatoid arthritis, breast pain, and menopause symptoms	Probably safe for short periods of time; may increase complications of pregnancy; may increase bleeding risk	Might be helpful for diabetic neuropathy; not helpful for eczema
Fenugreek	Historically used for digestive problems and to induce childbirth	Used in diabetics to stimulate breast milk production and used topically for wounds	May affect uterine contractions; may act like estrogen; side effects include diarrhea and worsening of asthma	May help to lower blood sugar; may increase milk production
Feverfew	Used for fevers, headaches, constipation, diarrhea, dizziness, and difficulty with labor	Currently used for migraine prevention, rheumatoid arthritis, psoriasis, and many other conditions	Side effects include nausea and digestive problems; may create rebound insomnia, headaches, anxiety, and myalgias if stopped after long-term use; may affect uterine contractions	AAN and AHS suggest it may be helpful for migraine prevention

Continued

TABLE 19.2 Dietary Supplements—cont'd

Name	Background	Common Use	Safety	Research
Garlic	Used for thousands of years across the world	Used for hypertension, hypercholesterolemia, and the common cold	Safe to use; side effects include breath/body odor and heartburn; may increase risk of bleeding	Conflicting evidence for hypercholesterolemia; may be helpful for hypertension; may reduce risk of GI cancers
Ginger	Used for thousands of years across the world	Used for nausea associated with surgery, motion, chemotherapy, or pregnancy; used for rheumatoid arthritis and osteoarthritis	Mild GI side effects; use with caution with individuals with gallstones; might interact with anticoagulants	May help nausea in pregnancy and chemotherapy
Ginkgo	One of the oldest living tree species; used in TCM for thousands of years	Used for dementia, eye problems, intermittent claudication, and tinnitus	Side effects include headache, stomachache, and allergic skin reactions; may interact with anticoagulants; raw ginkgo is poisonous	Ginkgo does not prevent or slow dementia or cognitive decline
Green tea	Used for medicinal purposes in China and Japan for thousands of years	Used to improve mental alertness, relieve digestive problems and headaches, and promote weight loss	Liver problems in small number of individuals taking concentrated extract; reduces serum levels of nadolol	Evidence that it enhances mental alertness; FDA-approved extract ointment for treating genital warts No consistent findings in cancer research
Horny goat weed	Used in TCM mainly for erectile dysfunction	Used for erectile dysfunction, osteoporosis, musculoskeletal pain, arthritis, and mental and physical fatigue	Long-term use may cause dizziness, vomiting, dry mouth, thirst, and nosebleed; may interact with antihypertensives and anticoagulants	There is no substantial research supporting the use
Lavender	Historically used to boost appetite and mood; also used for GI issues and anxiety	Used for anxiety, depression, intestinal problems, and pain	Topical use of diluted lavender oil is considered safe; not enough evidence for aromatherapy; may be poisonous if taken orally	Mixed results in studies for anxiety
Passion-flower	Native South Americans used it as a mild sedative	Used to treat anxiety and insomnia; also used for pain, attention-deficit/hyperactivity disorder	May cause drowsiness	Some evidence showing benefit in anxiety
Peppermint oil	Used for several thousands of years	Used for IBS, the common cold, headache, and muscle aches	Excessive doses can be toxic; side effects include allergic reactions and heartburn	May improve IBS symptoms; may relieve indigestion in combination with caraway oil; limited evidence for use with tension headaches
Rhodiola	Used in northern regions of Europe and Asia	Used for energy, stamina, and strength, to improve attention and memory, and to cope with stress		May enhance physical performance and reduce mental fatigue; may be as effective as sertraline in treating mild-moderate major depressive disorder
Soy	A plant in the pea family used for thousands of years in Asia	Used for menopausal symptoms, bone health, improving memory, and treating hypertension and hypercholesterolemia		May lower LDL levels if used to replace other proteins May help to reduce frequency and severity of hot flashes; does not slow bone loss in postmenopausal females
Thunder god vine	Grown in China and Taiwan and used for hundreds of years to treat inflammation	Currently used to treat autoimmune diseases, such as rheumatoid arthritis, multiple sclerosis, and systemic lupus erythematosus	Side effects include decreased bone mineral density, infertility, menstrual cycle changes, rash, diarrhea, headache, and hair loss; can be poisonous if not prepared properly	May improve rheumatoid arthritis symptoms and with greater benefit than sulfasalazine; combined with methotrexate, may provide synergistic benefit

TABLE 19.2 Dietary Supplements—cont'd

Name	Background	Common Use	Safety	Research
Turmeric	Related to ginger and grown primarily in India; used in Ayurvedic medicine to treat breathing problems, rheumatism, pain, and fatigue	Used for inflammation, arthritis, cancer, and multiple other conditions		Strong studies have not found the effectiveness of turmeric for inflammation; preliminary studies found that turmeric and ibuprofen control knee pain and reduce skin irritation after breast cancer radiation
Valerian	Used since early Greek and Roman times; Hippocrates documented its uses	Used for insomnia and anxiety mainly	Generally safe for short periods of time	There is inconsistent evidence of its use for insomnia; there is not enough evidence to demonstrate benefit for anxiety
Yohimbe	Native tree of Africa used as an aphrodisiac	Currently used for erectile dysfunction, athletic performance, weight loss, chest pain, hypertension, diabetic neuropathy	It has been associated with heart attacks and seizures It may cause tachycardia, anxiety, and hypertension	There is no substantial research demonstrating the benefits

AAN, American Academy of Neurology; AHS, American Headache Society; FDA, US Food and Drug Administration; GI, gastrointestinal; IBS, irritable bowel syndrome; LDL, low-density lipoprotein; MRI, magnetic resonance imaging; SCI, spinal cord injury; TCM, traditional Chinese medicine; UTIs, urinary tract infections.
Modified from the National Center for Complementary and Integrative Health Herb List

in the skin when exposed to ultraviolet light. Vitamin D is also absorbed from dietary sources of innate or fortified food. Research suggests that deficiency of vitamin D has been associated with poor health, diabetes, hyperglycemia, depression, muscle weakness, poor balance, and all-cause mortality.⁵⁰ Low vitamin D levels have also been associated with depression.^{4,59} Supplementation with vitamin D may be indicated in preventing falls in the elderly with low vitamin D levels.⁴⁰ It has also been associated with decreased pain, improved functional mobility and quality of life, and decreased mortality.^{9,10,104}

Glucosamine (amino sugar) and chondroitin sulfate (sulfated glycosaminoglycan) have had some controversy and mixed clinical reports regarding efficacy for the treatment of arthritis, and guidelines vary for use of these compounds in the setting of hip and knee arthritis.^{17,101,105}

Calcium and strontium contain nearly the same size atomic nucleus. As a result, strontium is readily taken up by bones and tooth enamel. Strontium ranelate is approved as a prescription drug for the treatment of osteoporosis in Europe and other countries but not in the United States; it is thought to be effective in both sexes.^{61,100,102} Additionally, it is found to be effective in the treatment of knee arthritis.⁹⁹ Strontium citrate is available in the United States as a nutritional supplement. Animal studies indicated that strontium citrate is taken up by bone in greater concentration than the ranelate form,¹²⁷ although no clinical trials have yet been conducted with the citrate form. The higher atomic number of strontium (38, compared with 20 for calcium) may create difficulty in the assessment of treatment efficacy based on dual-energy x-ray absorptiometry scans.⁷⁴

Essential Oils

Essential oils are concentrated from plants. They are mostly used aromatically or topically to improve physical, mental, or spiritual

well-being. There are many uses for essential oils that include stress and anxiety reduction, pain management, and improvement of nausea and vomiting. However, it is important to know that these products are not monitored by the US Food and Drug Administration. So, although most sources state that there are none to limited adverse effects of essential oils, there is no true regulation over the substances in each container.

Economics of Complementary and Integrative Health Care

It is well documented that rising costs are a tremendous challenge to health care in the United States. The Institute of Medicine found, in 2008, that the cost for treating pain in the United States alone was between \$560 and \$635 billion.⁵⁴ This number includes the cost of health care plus lost productivity. With this rising cost and the concern about health outcomes, systems are moving toward redesigning how health care is provided. In 2011, the Veterans Health Administration (VHA) created the Office of Patient Centered Care and Cultural Transformation with the goals of moving the VHA's focus on episodic care to continuous engagement throughout the lifetime of the individual and increasing access to integrative approaches. In 2020, the Department of Veterans Affairs (VA) published a compendium on use of CIH. In this document, the VA demonstrates an increase in use of CIH by over 70% between fiscal years 2017 and 2019.¹¹⁵

In 2009, Abbott Northwestern Hospital in Minneapolis, Minnesota, undertook a retrospective study to examine non-toxic, nonpharmacologic integrative approaches to its inpatient population. A total of 1839 inpatients, two-thirds of whom had never had CIH, underwent various CIH practices that included acupuncture, massage, reflexology (foot massage), music therapy, guided imagery, relaxation response stress reduction, and

aromatherapy. The researchers noted an immediate pain reduction of 55% with no side effects. When they analyzed the cost data, they found that when pain was reduced, they saved approximately \$2000 per stay, or approximately \$2 million. This was attributed to decreased medication use in those patients receiving integrative medicine.³⁶

Using CIH within a medical system may lead to economic cost benefits by reducing both direct and indirect costs of care. As such, it is important for future research to examine this area, especially in the United States where health care costs are a major challenge.

Summary

Integrative health care is accurately described as an approach to the delivery of health care, not as a separate entity in and of itself. In fact, many believe that integrative health care is the way all medical care, not just rehabilitation, will be in the future. Although it often incorporates many complementary practices, the key is a patient-centered integrative approach that includes all health and wellness practices that best serve each individual.

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